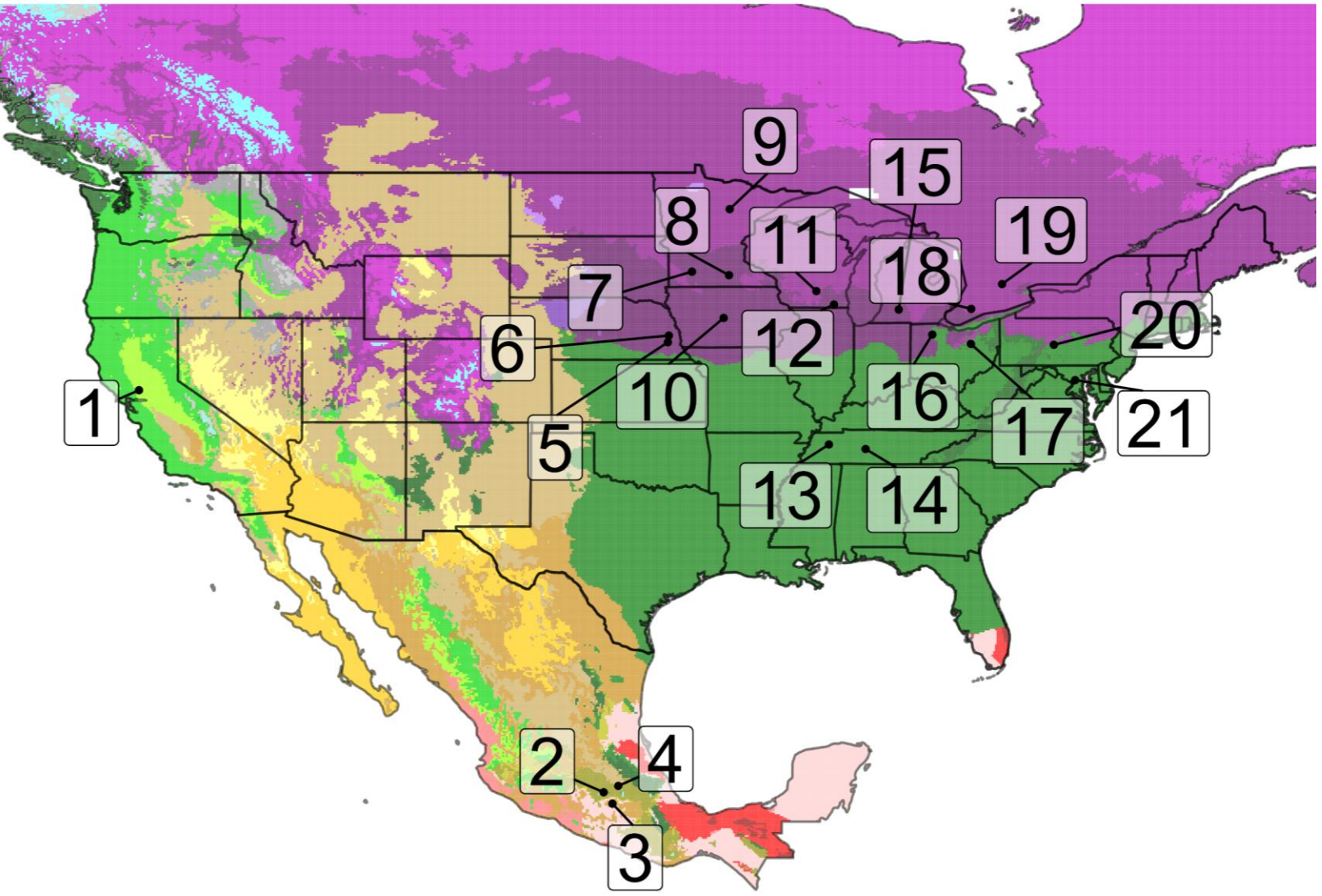
DRIVES example

DRIVES Sites

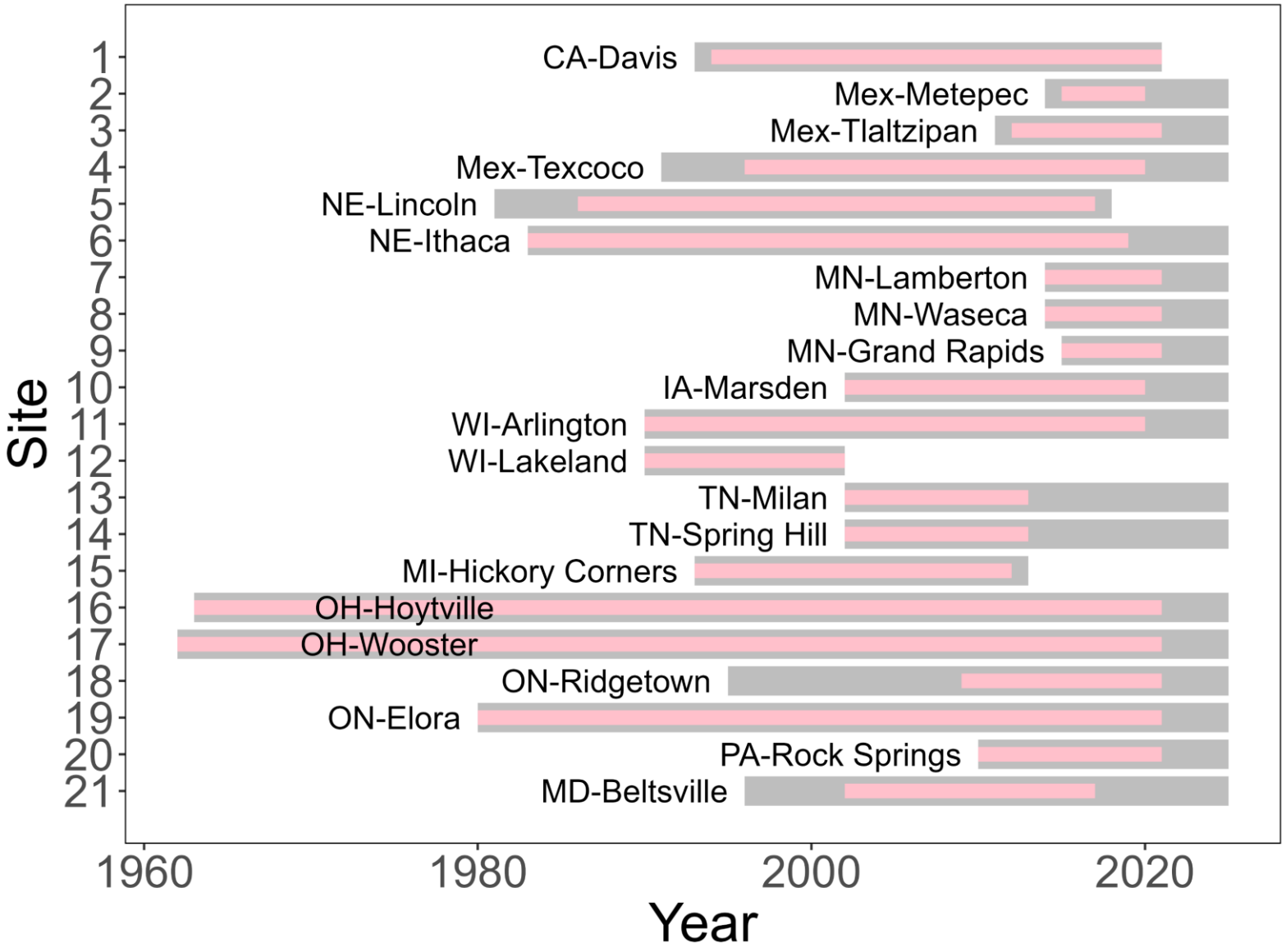


22 long-term experiments so far…

Criteria:

- At least 2 crop rotations
- At least 3 full cycles of rotations
- Every phase of rotation grown every year

DRIVES Data



21 long-term agricultural experiments

495 site-years of data

87,300 yield observations

311,600 daily weather observations

18 sites with long-term soils data

Why a Bayesian approach is helpful for DRIVES analysis

- Varying experimental designs and durations lead to uneven sample sizes across sites.
- We are interested in risk and uncertainty.

What are the effects of rotational complexity on productivity under varying growing conditions for **complete rotations** and their **component crops**?

Rotational complexity increases cropping system output under poorer growing conditions
Bybee-Finley et al. 2024, One Earth v7

Methods: Models

- Bayesian multi-level modelling
- Separate models used for each crop and for complete rotation
- Simplified model:

  - Output ($ ha^{-1}) = **a** + b*EI + **c***RCI + **d***(RCI × EI)

  - Random **intercepts** and **slopes** for management subgroups within sites

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    Crop rotations converted to continuous variable based on species richness and rotation length

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Methods: Models

- Bayesian multi-level modelling
- Separate models used for each crop and for complete rotation
- Simplified model:

Growing conditions quantified as average output of each management group-site-year

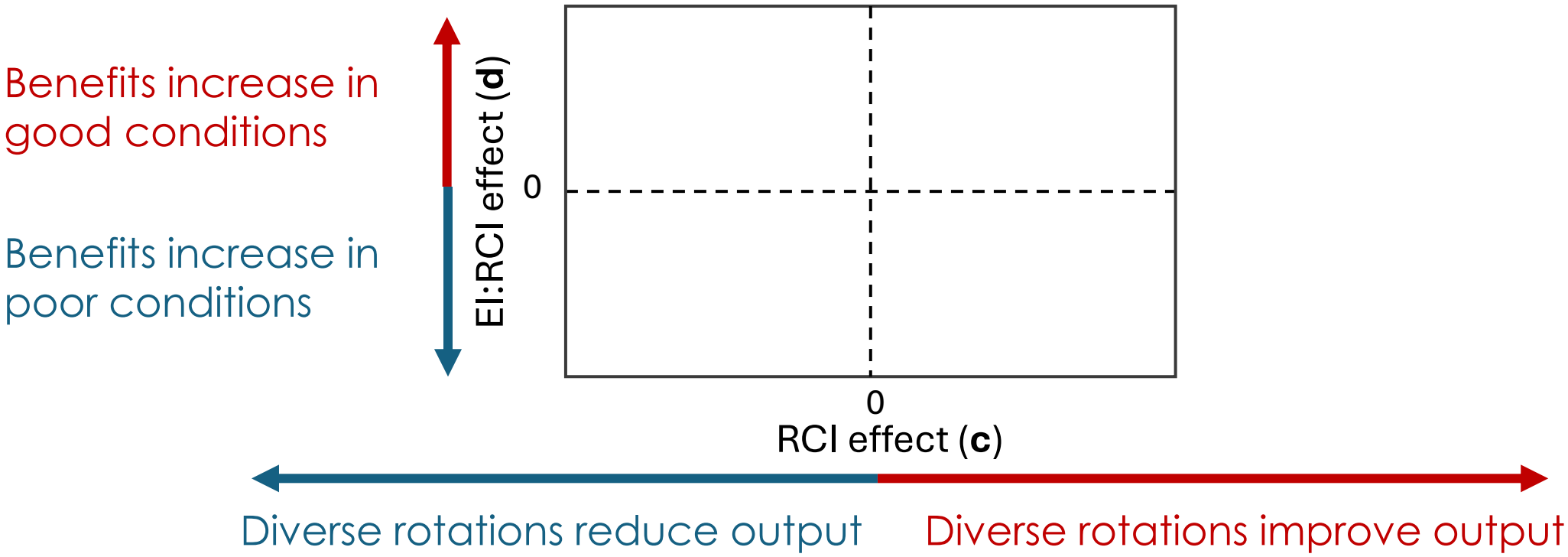
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Crop rotations converted to continuous variable based on species richness and rotation length

  - Random **intercepts** and **slopes** per management subgroup within site

Interactive effects of rotational complexity and growing conditions on output

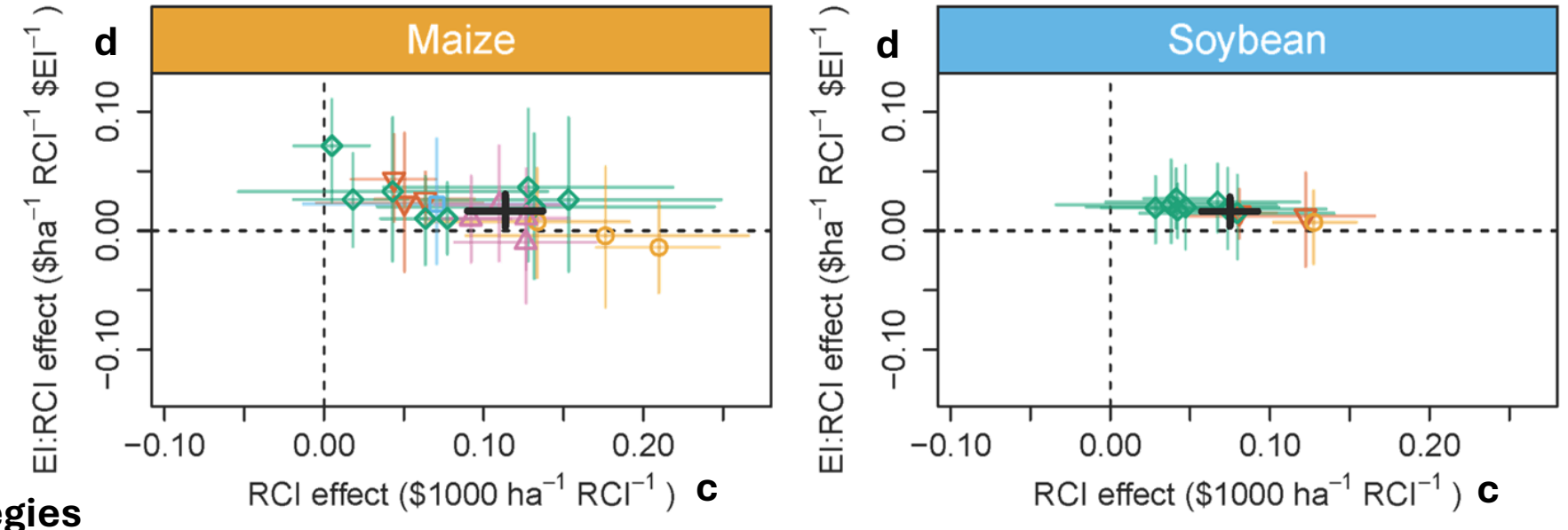
Output ($ ha^{-1}) = a + b*EI + RCI(**c** + **d***EI)



Individual crop results

Output ($ ha^{-1}) = a + b*EI + RCI(**c** + **d***EI)

Benefits were similar or greater in good conditions.



Diversification strategies

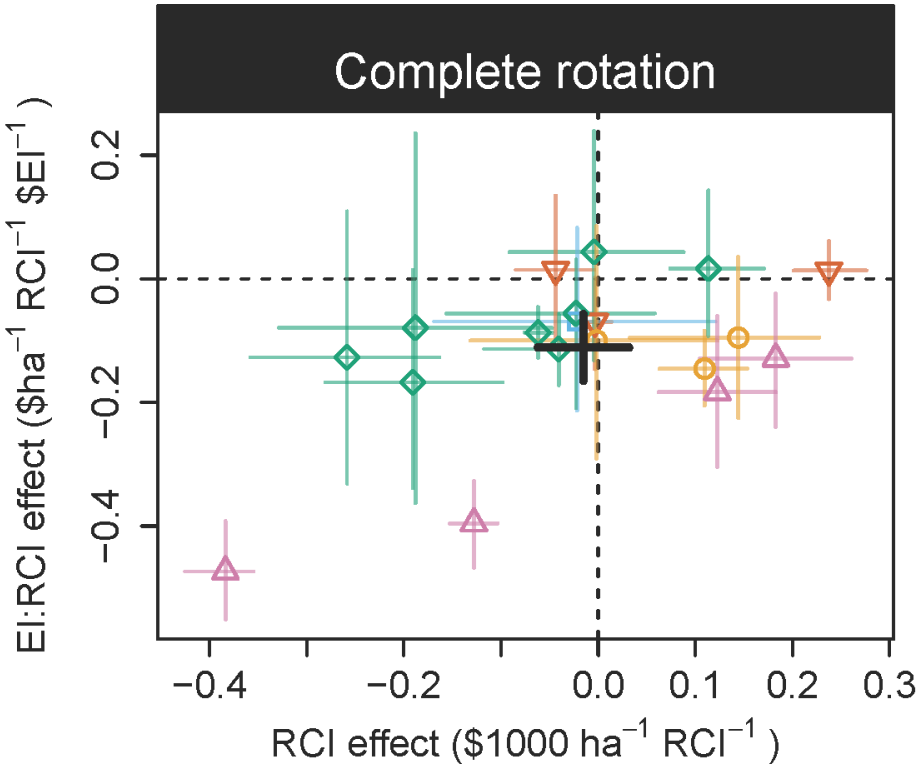
- Annual only
- Annual and cover
- Annual, cover and perennial
- Cover only
- Annual and perennial

Diverse rotations improved maize and soybean yield.

Complete rotation results

Output ($ ha^{-1}) = a + b*EI + RCI(**c** + **d***EI)

…but showed a trend of greater diversity benefits in poor conditions.



Evidence for risk mitigation!

Diversification strategies
- Annual only
- Annual and cover
- Annual, cover and perennial
- Cover only
- Annual and perennial

Diverse rotations had mixed outcomes for complete rotation output.

Average response to rotational complexity across 20 sites. A positive number means that output increased, and a negative number means that output decreased in more complex rotations. The 95% credibility interval indicates the range of responses supported by our data.

	Change in output per unit increase in RCI ($ ha^{-1})	95% credibility interval
Corn	+ 114	(+90 to +137)
Soybean	+ 75	(+57 to +92)
Spring small grain	+ 47	(-94 to +172)
Winter wheat	+ 4	(-13 to +22)
Complete rotation	- 15	(-62 to +32)

Code available on Ag data commons: https://doi.org/10.15482/USDA.ADC/25943899